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Belgium's SARS-CoV-2 contact tracing effectively curbed transmission, yet challenges persist in terms of delays, scalability and population adherence

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Abstract

Background Through contact tracing, researchers could reconstruct chains of linked cases by linking High-Risk Exposure Contacts (HREC) testing positive to their index case. We present chain-level descriptives, delays and an individual-level analysis of the events that determine chain progression.

Methods We used data from contact tracing outside of collectivities between September 2020 and March 2022 in Belgium. We present chain-level generation counts and structure. The individual-level analysis focusses on the events necessary to establish transmission chains from index cases: inclusion, successful contact with the call center, reporting HREC, identifying HREC and testing of HREC. We adjusted for age, sex, household, calendar time, region, vaccination, laboratory-confirmed prior infection, viral load, symptoms and the place in the transmission chain: we differentiated Cases Identified As HREC (CIAH, index cases that had been identified as HREC of another index case) from primary cases.

Results The percentage of CIAH over all cases was 14% and most CIAH (63%) were household-members of the index case. Unsuccessful contact of cases (34% of all index cases) and reporting no HREC (53% of successfully contacted index cases) ended most chains. Among identified HREC, 75% had at least one test result. Persons age 65 and over were less likely successfully contacted (OR 0.59 (95%CI 0.58–0.59)), reporting HREC (OR 0.55 (95%CI 0.54–0.56)) or tested as HREC (OR 0.87 (95%CI 0.85–0.89), reference 25–44 year old). Compared to primary cases, CIAH were associated with less HREC and lower HREC test positivity. Delays however were comparable. The delay from symptom-onset to test result exceeded 4 days for 29% of CIAH.

Conclusions While contact tracing effectively reduced transmission within identified chains, over 80% of cases remained unlinked due to, amongst others, contact failures and non-reporting of HREC. Future efforts would benefit from operational improvements and enhanced population participation.

Keywords Contact tracing, SARS-CoV-2, Public health impact, Evaluation, COVID-19, Serial interval, High-risk contacts

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Introduction

The large-scale contact tracing discussed in this paper was implemented in Belgium at the end of the first lockdown, May 2020, until March 2022. Together with other non-pharmaceutical interventions (NPI), it was seen as a means to keep the impact of COVID-19 within hospital capacity [1].

Contact tracing was initiated by a positive laboratory test without a recent (<60–90 days) history of SARS-CoV-2 infection in a community-dwelling person. Contact tracing within firms, schools and large collectivities (a collectivity was defined within the housing context as a group of persons living in the same building) was separate from general contact tracing. A person with a laboratory-confirmed infection, referred to as index case, was contacted and interviewed about contacts. For symptomatic and asymptomatic index cases the period of interest started two days before symptom-onset and first positive test respectively. A High-Risk Exposure Contact (HREC) was defined as anyone with whom the index case had a close encounter (e.g. physical contact or contact at <1.5 m without the correct use of face masks for over 15 min) [2]. Index cases were reminded about standard practice (self-isolation, handwashing, ...) and contacts were required to get PCR-tests and go into quarantine. To reduce the opportunity cost and allow for flexible upscaling, interviews were performed by call agents working at call centers outside of the existing public health system.

Comparable contact tracing systems were widely implemented internationally, but the evaluation of their impact is hindered by the absence of specific contact tracing parameters [3] and a lack of details in published literature on the implementation of contact tracing activities [4]. Proposed effectiveness indicators include operational metrics such as tracing delays and primary coverage (inclusion of cases). Effectiveness can also be evaluated by epidemiological indicators such as case-finding and incidence as well as by more indirect indicators such as secondary coverage (the proportion of cases identified as contacts) or cases identified before symptom-onset [5]. Finally, serial intervals, the interval between the symptom onsets of the index case and HREC, are considered a possible parameter to evaluate contact tracing's effectiveness. Successful contact tracing, and case isolation in particular, should shorten the serial interval [6]. A relatively small set of empirical, observational studies, reporting on these indicators have concluded that contact tracing can be effective [7, 8]: four (66.7%) out of six COVID-19 studies, found contact tracing interventions were associated with improvements in at least one of the aforementioned indicators [7]. The small amount of empirical data, the variety of ways in which both testing and contact tracing were defined and the different measures for quantifying

transmission reduction have made it difficult to inform optimal strategies for contact tracing [7].

In comparison to researchers from other countries, Belgian researchers benefitted from a centralized database with high inclusion of cases as index cases (>90%) and high PCR-test coverage among HREC, as testing was mandatory during most of the period in which contact tracing was performed. Mandatory testing of HREC ended in January 2022 during a period of high incidence associated with the Omicron variant. The centralized database can be linked to the vaccination registry, vaccine roll-out started in December 2020, and to an exhaustive database of laboratory tests and prescriptions. Previous studies have reported that, except for periods with very high incidences in the fall of 2020 and 2021, performance was maintained from mid-2020 to early 2022. Turn-around-times in general were short, 1 day between test result and contact with the call center. The main delay was the period between symptom-onset and healthcare seeking behavior. For a large part of 2021, they reported a relatively high average secondary coverage of 22% and effective case finding: detecting up to 24% of new cases while only performing 6% of the tests [9]. Belgian contact tracing has been reported to reduce transmission: cases known to contact tracing reduced their number of contacts and were associated with lower secondary attack rates and shorter serial intervals [10].

In this paper we explored the Belgian contact tracing data in depth, motivated by the counterintuitive finding that contact tracing revealed predominantly isolated cases rather than extensive transmission networks. Through an analysis of events underlying case linkage and person-level regression analysis of endpoints and delays, we discuss how disease, design and adherence aspects impacted contact tracing.

Methods

Data

We included national data from September 2020 (2020-09-01), when the centralized collection of contact tracing data was in its final form, to the end of February 2022 (2022-02-28), when contact tracing was officially ended. Pseudonymized data from contact tracing was linked to exhaustive laboratory testing data on PCR and antigen (AG)-tests, prescription data and exhaustive data on vaccination (SARS-CoV-2 vaccination registry). The variables included in this study were personal characteristics (age at the time of contact with the call center, sex as in the population registry, region of residence), contact tracing data (dates of contact with call center, results of the call, type of contact (e.g. household), self-reported symptoms and symptom-onset), test and prescription-data (self-reported symptom-onset, type of test, date of

testing, result and date of result, viral load) and vaccination data (number of doses and date of administration).

Chains and delays

We identified six events necessary for case linkage: (1) including cases in contact tracing as index cases, (2) successful contact between index case and call center, (3) contacted index case reported HREC, (4) HREC were identified, (5) linking a test to an identified HREC and (6) test positivity. Possible chain endpoints are associated with these events: No inclusion (ending at event 1) unsuccessful contact (ending at event 2) no HREC reported (ending at event 3), the reported HREC could not be identified (ending at event 4), no tests were found for the identified HREC (ending at event 5) and finally all HREC tested negative in the -1:20 days following contact with the call center (ending at event 6). Only when all events were successful cases could be linked.

- (1) Contact tracing operated through a ticketing system to organize its tasks. When a test came back positive for a community-dwelling person without recent (60–90 days) positive test, a ticket was made. If no ticket was made in the 20 days following the positive test, the case was not included in contact tracing. This was linked to administrative problems or technical errors.
- (2) Once the ticket was made, the call center tried to telephonically contact the index case. If telephonic contact could not be made, a field agent directly contacted the index case. If contact was made and the index case participated, the contact was successful. Some cases did not participate and during periods of high incidence not all cases were contacted.
- (3) An index case reported contacts, whereabouts and collectivities. Based on the characteristics of the contact, the call center agent labelled the contacts high (HREC) or low risk. Contact tracing within the collectivity was done by the medical professionals associated with the collectivity and tracing details were not included in centralized contact tracing. Cases detected through contact tracing in collectivities were however again included in centralized contact tracing to perform tracing of possible contacts outside of the collectivity.
- (4) While an index case would report contact details for their HREC, HREC could only be formally identified by their identity-number (INSZ) (reported by the index or self-reported during a telephone interview by the HREC) or if their test prescription code was reported together with the test results by the laboratory.

(5) Identified HREC could be linked to the exhaustive database of laboratory results. Tests from -1 to 20 days after contact with the call center were considered tests linked to contact tracing. We included tests at day -1, one day before contact with the call center, as these tests had consequences for testing associated with contact tracing. HREC tested at day -1 would typically still be contacted by the call center but, if the test was positive, further testing was not required. Positive tests from earlier dates (more than 1 day before contact with the call center) were not possible as the HREC would then already be included as index case and excluded as possible HREC.

- (6) A HREC was only considered negative if all tests within the -1:20 day period were negative.

Among index cases, we differentiated Cases previously (-20:-1 days) Identified As HREC (CIAH) from primary cases. CIAH could be further subdivided by their generation in the chain of linked cases (secondary/tertiary/...). Some cases were contacted as index case and HREC on the same day. Such cases with simultaneous ascertainment were labelled cluster cases. A subgroup of cluster cases were loop cases. If case A reported case B as HREC while simultaneously case B reported case A as HREC, a putative transmission chain would loop. We first describe the proportion of cases linked and the generational counts (number of primary, secondary, ... index cases). We then classified the index cases by the furthest endpoint reached in their transmission chains. For example, an index case that reported two identified HREC of which one remained untested while the other tested negative would be classified as "HREC tested negative" (the furthest endpoint reached). Loop cases were excluded from endpoint classification analysis due to their circular transmission structure. We illustrated these definitions and endpoints in Fig. 1.

We present average delays associated with contact tracing. We included (A) date of symptom-onset, (B) date of testing, (C) date of test result, (D) date of contact with the call center. We included these delays for both index cases and HREC and compared primary cases to CIAH. Delays were calculated from dates in number of days as time of day was unavailable.

Regression models

To estimate the association of several parameters with successfully contacting the index case (event 2), reporting HREC (event 3) and finding test results for a HREC (event 5), we fitted mixed logistic regression models. Test positivity (event 6) was a topic of previous research [11]. In each regression model, age group (6 levels: 0–5, 6–17, 18–24, 25–44, 45–64, 65 + years), sex (binary: Male

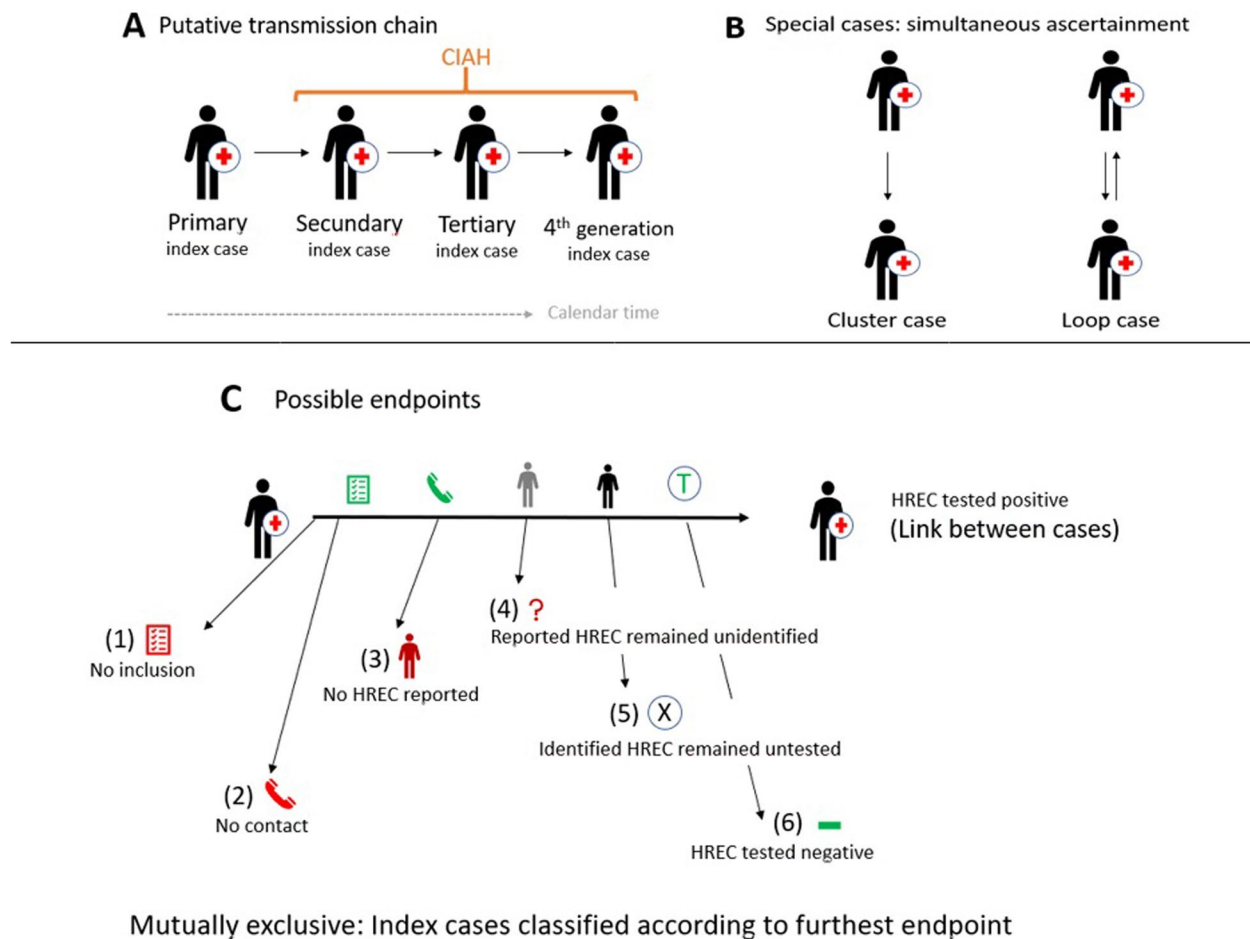


Fig. 1 Overview of the different index cases (**A** and **B**) and the possible endpoints for case linkage (**C**). The possible endpoints represent instances where an event was not successfully completed (CIAH: cases identified as high risk exposure contact)

and Female) and region (3 levels: Flanders, Wallonia and Brussels) were included as fixed factor effects and calendar week (50 levels) was included as a random intercept. Model building was based on Akaike's Information Criterion (AIC), favoring a parsimonious model. Regression models were fitted only to 2021 data. Several variables were not collected until early 2021 (e.g. viral load). In 2022, several changes were made to the contact tracing procedure (e.g. testing of HREC no longer mandatory if asymptomatic).

Successful contact between index case and call center

We compared contacted index cases to those not successfully contacted. No data was available on why the index was not successfully contacted (e.g. contact tracing did not reach out, index could not be reached or declined participation).

$$\begin{aligned}
 P(\text{contact}_{index}) \sim & \text{age}_{index} \\
 & + \text{sex}_{index} + \text{region}_{index} + \text{calendar week} \\
 & + \text{symptoms}_{index} + \text{high.viral.load}_{index} \\
 & + \text{primary|CIAH} + \text{vaccination}_{index} \\
 & + \text{prior.infection}_{index}
 \end{aligned}$$

Symptoms (binary: symptomatic or asymptomatic) was based on self-reported information on date of symptom-onset obtained from the prescription form. *High.viral.load* (binary) indicated the presence or absence of high viral load ($\geq 10^7$ RNA copies/mL). *primary|CIAH* (binary) referred to primary case or CIAH. *Vaccination* (binary) defined if a person had received at least one vaccination dose or not. *Prior.infection* (binary) defined if a person had a laboratory-confirmed (PCR or AG-test) infection 60 days or more before inclusion in contact tracing or not.

Contacted index reported HREC

Given successful contact, an index might report HREC or not.

$$\begin{aligned}
 P(\text{reporting}_{index}) &\sim \text{age}_{index} \\
 &+ \text{sex}_{index} + \text{region}_{index} \\
 &+ \text{calendar week} + \text{symptoms}_{index} \\
 &+ \text{high.viral.load}_{index} + \text{primary|CIAH} \\
 &+ \text{vaccination}_{index} + \text{prior infection}_{index}
 \end{aligned}$$

symptoms_{index} (binary) now also included, in addition to the information from the prescription form, information obtained during contact tracing (interview) on the presence or absence of symptoms.

Linking a test to an identified HREC

Could a test be linked to an identified HREC or not.

$$\begin{aligned}
 P(\text{test}_{HREC}) &\sim \text{age}_{HREC} \\
 &+ \text{sex}_{HREC} + \text{region}_{HREC} + \text{calendar week} \\
 &+ (\text{household})_{index,HREC} + \text{symptoms}_{HREC} \\
 &+ \text{vaccination}_{HREC} + \text{prior infection}_{HREC}
 \end{aligned}$$

$(\text{household})_{index,HREC}$ (binary) identified HREC that were household members of the index case. symptoms_{HREC} (binary) referred to the presence or absence of self-reported symptoms during the telephone interview with the HREC.

Results

Events and linkage

Chain descriptives

Over the study period, 3 451 461 cases were reported. Of these, 2 830 198 (82%) were primary cases, 483 205 (14%, secondary coverage) were CIAH and 4% were not included in contact tracing. While the proportion of CIAH over all cases was 14% over the complete period of contact tracing, this proportion was notably higher during the first half of 2021 when 25% of cases were CIAH (18% during the second half of 2021, 9% in 2022 and 8% in 2020). The secondary coverage translated into short transmission chains: we observed only 93 444 third generation index cases and 9 778 fourth (and higher) generation index cases (Fig. 2, upper). We present an overview of chains by time period in Supplementary Fig. 1. The percentage of contacted index cases that remained unlinked with another case was 85% over the complete study period and 81–82% in 2021. Those indexes that could be linked with further cases had, on average, 1.5 CIAH.

We investigated why such a high proportion of index cases remained unlinked. Notable final endpoints for index case were: 34% were not contacted successfully. This percentage was highest in early 2022 and was lowest in the first half of 2021 (Table 1). 53% of successfully contacted index cases reported no HREC. For those

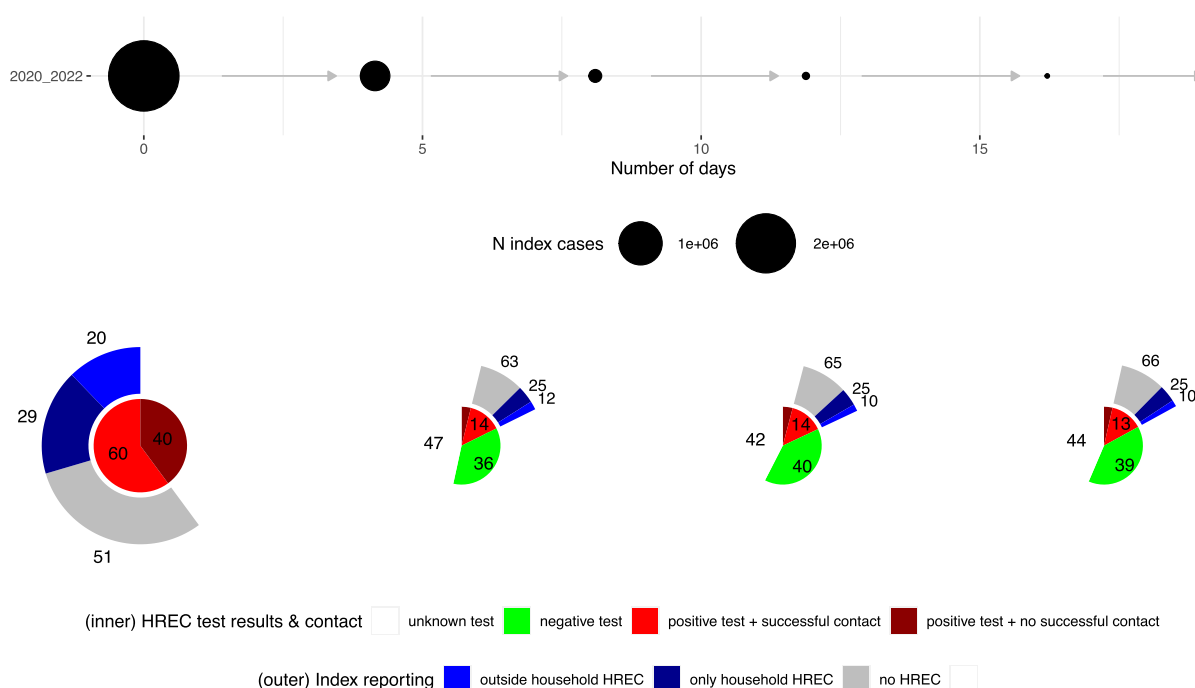


Fig. 2 (Upper) Number of cases by generation (left to right, up until the fifth generation). The distance between the nodes represents the average serial interval. (Lower) Proportions by generation (left to right, up until the fourth generation). (Inner circle) test and contact with call center results, (Outer circle) proportion of index cases reporting (household) HREC. In this graph, the cases that were not included in contact tracing (no ticket) were presented as no successful contact, Belgium, 2020-09-01 to 2022-02-28. (N included index cases: 2,830,198 (primary), 379,983 (secondary), 93,444 (tertiary) and 9,778 (fourth generation and higher), N included HREC 2,754,985 which refers to the HREC tickets as a large part remained unidentified)

Table 1 Overview of proportions by period and by primary/CIAH index case of the final end points of index cases (CIAH: cases identified as High-Risk Exposure Contacts (CIAH)), Belgium, 2020-09-01 to 2022-02-28. (N included all = 3,451,461, primary = 2,830,198 and CIAH = 469,192, loop cases are withheld from the analysis as they do not have a final endpoint, leading to a smaller number for the included CIAH. Other numbers are provided in Supplementary Table 2)

Variable	Group	2020_2 (N=564,126)	2021_1 (N=425,137)	2021_2 (1,032,704)	2022_1 (N=1,429,494)	2020_2022 (N=3,451,461)	Primary (N=2,830,198)	CIAH (N=469,192)
(1) Not included	Among cases	0.09	0.05	0.02	0.03	0.04		
(2) No contact	Given (1) inclusion	0.18	0.05	0.30	0.52	0.34	0.37	0.19
(3) No HREC reported	Given (2) contact	0.57	0.53	0.48	0.56	0.53	0.51	0.64
(4) HREC reported, not identified		0.20	0.06	0.07	0.12	0.11	0.12	0.07
(5) HREC identified, not tested		0.04	0.02	0.03	0.12	0.06	0.06	0.05
(6) HREC tested negative		0.11	0.22	0.23	0.07	0.16	0.16	0.14
(6) HREC tested positive		0.08	0.18	0.18	0.14	0.15	0.15	0.11

that did report HREC, 17% had only HREC without test results (11% remained unidentified, 6% were identified but not tested) and for 16% of index cases all HREC tested negative. In Fig. 2 (lower) the endpoints of transmission chains are summarized from the perspective of the HREC. A minority of HREC remained untested, 45% of all HREC, (25% of the identified HREC) and 37% (of all HREC) tested negative. We observed multiple differences between primary cases and HREC testing positive (= CIAH). CIAH were more often successfully contacted as index case by the contact center compared to primary cases; 81% vs 62% (given inclusion). Given successful contact, CIAH reported HREC less often compared to primary cases: 37% vs 49%, respectively. For CIAH who reported HREC, the average number was lower compared to primary cases: 2.7 and 2.3. The lower number of HREC concerned both household and non-household contacts. Positivity rates of HREC were only slightly lower for CIAH (36% vs 37%). The overall temporal variation in positivity rate was much larger (Supplementary Fig. 2). Other differences, such as the younger age of CIAH and the higher proportion of cases with a laboratory-confirmed prior infection among primary cases are presented in Table 2.

Putative transmission chains often reflected possible household transmission. Most HREC (63%) were household contacts. This is further illustrated by the age profiles of HREC and CIAH. Age density curves for HREC and CIAH have bimodal peaks at 10–12 years and 40 years (Supplementary Fig. 3). The secondary coverage was highest in the age group of 6 to 18 years of age and lowest in those aged 65 years and over (Supplementary Fig. 4).

The proportion of cluster cases among CIAH increased to 13% in the second half of 2021 (with 6% of loop cases).

Over the complete study period, cluster cases were less prevalent: 9% cluster cases and 3% loop cases (Supplementary Table 1). Serial intervals were 4.02 days on average with 8% and 16% of intervals negative and longer than 7 days respectively. Serial intervals were shorter during the second half of 2021 (3.96 days) and the first half of 2022 (3.86 days) (Supplementary Table 1 & Fig. 1). The serial interval contracted from 4.15 days for primary cases to 3.93 days for CIAH (Table 2 and density plots in Supplementary Fig. 5).

Regression models

Over the study period there was an upward trend in the proportion of cases included. From 10 January 2022 onwards testing of asymptomatic HREC was no longer mandatory and this is reflected in a decrease in the proportion of identified HREC with a test. Other trends were linked to changes in case incidence. Notably during the fall of 2020, the high incidence resulted in less identification and testing of HREC, where high incidences later in the pandemic had a more upstream effect; reduced contact with index cases (Fig. 3 (unadjusted proportions over calendar time) and Supplementary Fig. 6 (adjusted, random intercept estimates)).

The age group 6 to 17 years of age was most often successfully contacted (OR 1.49, 95%CI 1.47–1.52) and most likely to report HREC (OR 1.33, 95%CI 1.31–1.34). Persons aged 65 years and over were less likely to be successfully contacted (OR 0.59 (95%CI 0.58–0.59)), report HREC (OR 0.55 (95%CI 0.54–0.56)) or get a test as HREC (OR 0.87 (95%CI 0.85–0.89)) compared to persons aged 25–44 years. The youngest age group, under 6 years, was least likely to get tested after being identified as HREC (OR 0.22, 95%CI 0.22–0.23). Males were less likely successfully contacted (OR 0.92, 95%CI 0.91–0.93),

Table 2 Comparison between index cases: primary cases and CIAH, with respect to personal characteristics, HREC and serial intervals. For proportions an exact 95% confidence interval was calculated using Wilson's score method, for continuous variables the 95% confidence interval was based on normal approximations, Belgium, 2020-09-01 to 2022-02-28

Group	Variable	Primary (N=2 830 198)	CIAH (N=483 205)
Age (in years)	Average	37.34 (37.32–37.36)	32.61 (32.54–32.68)
	Median	35	32
Sex	Female	0.54 (0.540–0.541)	0.52 (0.519–0.521)
History of infection	Prior infection	0.06 (0.06–0.06)	0.04 (0.039–0.040)
Vaccination	At least one	0.49 (0.489–0.491)	0.42 (0.419–0.421)
	Recent (< 90 days)	0.17 (0.170–0.170)	0.15 (0.149–0.151)
Viral load	High	0.24 (0.240–0.240)	0.20 (0.199–0.201)
Symptoms	Reported	0.64 (0.639–0.641)	0.65 (0.649–0.651)
HREC	Total number	2 432 556	322 429
	Average per index	2.74 (2.737–2.743)	2.39 (2.382–2.398)
	Average number of household contacts	1.55 (1.548–1.552)	1.45 (1.446–1.455)
	Average number of contacts outside the household	1.19 (1.186–1.194)	0.94 (0.934–0.946)
HREC test	Number of HREC tested	1 295 153	184 447
	Positivity rate	0.37 (0.369–0.371)	0.36 (0.358–0.362)
	Positivity rate (household)	0.43 (0.429–0.431)	0.41 (0.407–0.413)
	Positivity rate (non-household)	0.23 (0.229–0.231)	0.20 (0.197–0.203)
Serial Interval	Number	236 135	25 669
	Average	4.15 (4.14–4.16)	3.93 (3.90–3.97)

reporting HREC (OR 0.92, 95%CI 0.91–0.93) or tested as HREC (OR 0.96, 95%CI 0.95–0.97) compared to females.

Persons with a high viral load were more likely successfully contacted (OR 1.85, 95%CI 1.81–1.89) and more likely to report HREC (OR 1.16, 95%CI 1.14–1.18, reference: no high viral load). CIAH were more likely successfully contacted (OR 1.84, 95%CI 1.82–1.87), but less likely to report HREC (OR 0.52, 95%CI 0.52–0.53, reference: primary cases). Self-reporting symptoms was associated with successful contact (OR 1.20, 95%CI 1.18–1.21), reporting HREC (OR 2.6, 95%CI 2.58–2.62) and getting a test as HREC (OR 1.07, 95%CI 1.05–1.09, reference: asymptomatic). If a HREC shared a household with the index, the HREC was less likely to get tested (OR 0.9 95%CI 0.88–0.91, reference: not-household HREC).

Finally, vaccination was associated with decreased odds of successful contact (OR 0.90 95%CI 0.88–0.91), increased odds of reporting HREC (OR 1.22 95%CI 1.21–1.23) and testing as HREC (OR 1.49, 95%CI 1.46–1.51) compared to unvaccinated persons (Table 3).

Delays

The delays are presented in Fig. 4. In an effort to identify CIAH early, a two test strategy was in place from 2021 onwards: if the first test (as soon as possible after contact with the call center) was negative, a second test needed to be done 7 days after the last high-risk exposure or whenever symptoms occurred. One quarter (26%) of CIAH first had a negative test before eventually testing positive. Even if CIAH were symptomatic at the time of testing, 10% had a negative test before testing positive with a second test. In 2020 this strategy was not in place and typically only one test was performed. The first positive test then preceded the first test in general, indicating that positive HREC were getting a test faster than required by contact tracing (Supplementary Fig. 7). Testing could also precede contact with the call center: 10% of all HREC (9% of asymptomatic HREC) were tested on the day before contact, 30% were tested on the day of contact (29% of asymptomatic HREC).

On average, the contact with the call center did precede the symptom-onset of CIAH, but 43% reported a symptom-onset before being contacted by the call center. When comparing CIAH to primary cases, delays were comparable. For 30% of primary cases and 29% of CIAH the delay between symptom-onset and test result was four days or more. The time interval from symptom-onset to contact with the call center as index case on average was 3.5 days for CIAH and 3.7 days for primary cases. This delay was 4 days or more for 46% of primary cases and 41% of CIAH. The shortest delay in the process was typically from test to test result, 0.6 days. There were 0.7 days on average from test result to contact with call center. Both delays were longer in 2020, around 1 day each. In addition, in 2020, 50% of CIAH were symptomatic before they were contacted.

Discussion

Belgium's COVID-19 contact tracing included millions of cases and contacts, most cases however remained isolated finds. The high proportion of unlinked cases (> 80%) reflects the cumulative effect of failures at multiple contact tracing stages, primarily unsuccessful contact with index cases and failure to report HREC among those contacted. While we demonstrated contact tracing effectiveness for cases within the system, the majority of infections remained 'isolated finds' without clear epidemiological links. This illustrates an important limitation of contact tracing as a comprehensive surveillance

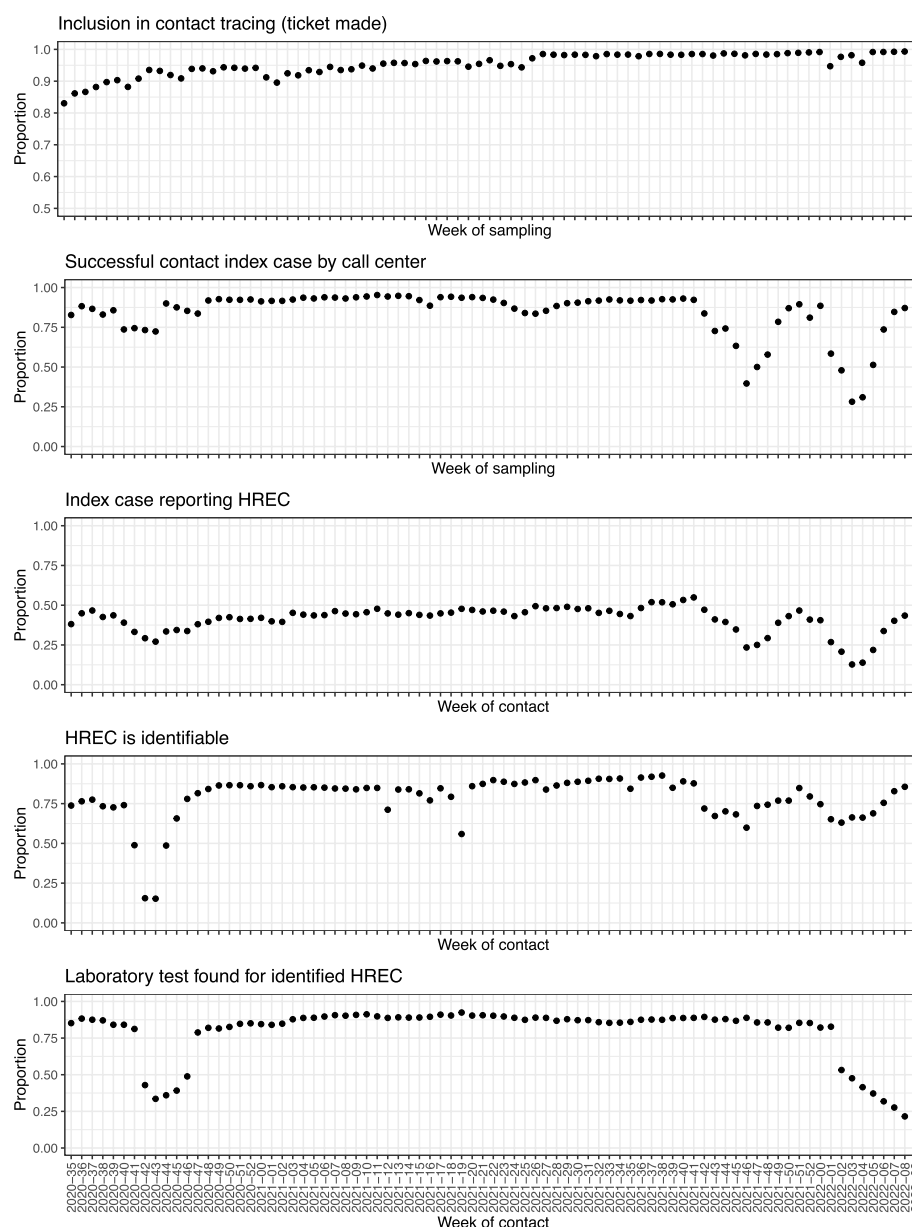


Fig. 3 Overview of the unadjusted proportion per week for Inclusion in contact tracing (ticket made), Successful contact index case with call center, Index case reporting HREC, HREC is identifiable, laboratory test found for HREC, Belgium, 2020-09-01 to 2022-02-28 (N included: 1e row (all cases) 3,451,461, 2e row (included cases) 3,312,495, 3e row (contacted cases) 2,178,819, 4e row (all HREC) 2,754,985, 5e row (identified HREC) 2,231,538)

tool once disease becomes endemic and widespread—it may effectively interrupt identifiable transmission chains while missing the majority of transmission occurring outside the observable network. For those identified chains, a proportion of delays were too long to fully interrupt transmission from CIAH. Before discussing these findings further, context on overall case-finding and the connection between contact tracing’s design and disease transmission should be explored.

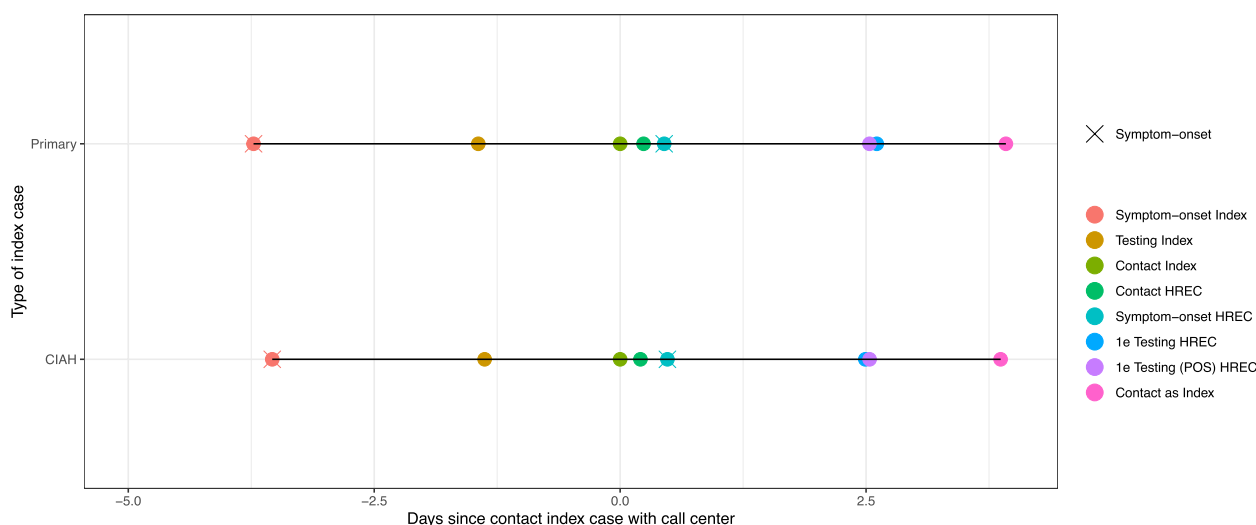
Countries that successfully controlled smaller outbreaks through, amongst others, contact tracing, have reported few to no undetected cases [12]. Primary

coverage should at least be 70% of symptomatic index cases [13, 14]. Belgian contact tracing exceeded this threshold. Contact tracing could however also be considered when case-finding is low as contact tracing itself is a performant case-finding tool. Contact tracing’s positivity rates, around 30%, were higher than those associated with other reasons for testing (e.g. travel, symptoms, ...) [15, 16]. Case-finding in itself already has considerable impact on disease transmission [17, 18].

Transmission of SARS-CoV-2 has been linked to three C’s (crowding, closed spaces, close contacts) and a potential for superspreading [19]. Given the NPIs, a ventilation

Table 3 Odds Ratios and 95% Confidence Intervals (95% CI) obtained from the mixed logistic regression for successful contact index case, index case reporting HREC and laboratory test found for HREC, Belgium, 2020-09-01 to 2022-02-28

Variables	Values	Odds Ratio Contact	95% CI Contact	Odds Ratio Reporting	95% CI Reporting	Odds Ratio HREC test	95% CI HREC test
Intercept	Intercept	25.22	(20.25–31.41) *	0.64	(0.61–0.68) *	7.85	(7.21–8.54) *
Age Group	[25,45)	ref					
	[00,6)	1.49	(1.44–1.53) *	1.21	(1.18–1.23) *	0.22	(0.22–0.23) *
	[6,18)	1.49	(1.47–1.52) *	1.33	(1.31–1.34) *	0.93	(0.92–0.95) *
	[18,25)	0.86	(0.84–0.87) *	1.12	(1.1–1.13) *	0.86	(0.84–0.88) *
	[45,65)	0.93	(0.92–0.94) *	0.83	(0.82–0.84) *	1.07	(1.05–1.08) *
	[65,Inf)	0.59	(0.58–0.59) *	0.55	(0.54–0.56) *	0.87	(0.85–0.89) *
Sex	F	ref					
	M	0.92	(0.91–0.93) *	0.92	(0.91–0.93) *	0.96	(0.95–0.97) *
Region	Brussels-Capital	ref					
	Flanders	0.35	(0.34–0.35) *	1.13	(1.12–1.15) *	1.12	(1.1–1.14) *
	Wallonia	0.57	(0.56–0.58) *	0.68	(0.67–0.69) *	0.91	(0.89–0.93) *
Viral load	No high viral load	ref					
	High viral load	1.85	(1.81–1.89) *	1.16	(1.14–1.18) *		
	Unknown viral load	0.67	(0.66–0.67) *	0.89	(0.88–0.9) *		
Index	Primary	ref					
	CIAH	1.84	(1.82–1.87) *	0.52	(0.52–0.53) *		
Vaccination	Unvaccinated	ref					
	Vaccinated	0.9	(0.88–0.91) *	1.22	(1.21–1.23) *	1.49	(1.46–1.51) *
Prior infection	No prior infection	ref					
	Prior infection	0.59	(0.58–0.61) *	1.03	(1.01–1.06) *	1.17	(1.16–1.19) *
Symptoms	Asymptomatic	ref					
	Symptomatic	1.2	(1.18–1.21) *	2.6	(2.58–2.62) *	1.07	(1.05–1.09) *
Household	Outside household	ref					
	Household					0.9	(0.88–0.91) *

**Fig. 4** Delays over the study period by primary cases (upper) and CIAH (lower). The analysis is centered around the day of contact between index case and the call center. Delays to the left concern the index case while delays to the right concern the HREC, Belgium, 2020-09-01 to 2022-02-28. (Delays were calculated separately. There is more uncertainty around delays involving symptom-onset as the numbers included were lower, uncertainty measures and N can be found in Supplementary Table 3)

plan and the presence of separate contact tracing in collectivities (hospitals, nursing homes, schools), centralized contact tracing in Belgium focused on what was possibly one of the main remaining routes of transmission:

high-risk exposure of known contacts. Our current paper indirectly investigated the assumption that this route was indeed one of the remaining contributors to infections by examining evidence of the reporting index case

as the source of infection of the HREC. We observed serial intervals with a length close to what has previously been reported by a systematic review [20] and a limited amount of cluster cases. Previous research already indicated that index case characteristics were significant when fitting the test positivity of HREC [11]. CIAH were associated with lower test positivity of their HREC (OR 0.93 (95%CI 0.91–0.96)) [21]. However, background incidence also remained significantly associated with HREC test positivity. High-risk exposure as defined in contact tracing was therefore an established, but not the sole remaining route. Quantifying this route in a context of unclear additional risk factors such as airborne transmission [22] is an important topic for future research.

Another aspect of the HREC definition was the start point of the contact tracing window (2 days prior to symptom-onset or positive test (whichever was first)). A Belgian study in university students investigated expanding the period to 3–4 days prior [23]. They observed that there was a mean difference of 1.8 days in symptom-onset dates between an index case and its HREC. This is much shorter than the average serial interval reported in this study. They concluded that such a design would lead to the inclusion of a high fraction of sibling cases in the traced group, a concept related to our cluster indexes.

While we described Belgium's secondary contact tracing coverage as limited, the percentage is within the range reported by other countries: e.g. US, 5.8% [24] and Catalonia, Spain, 34–58% [25]. In addition, our two main end points: the failure to contact index cases and, among contacted indexes, not reporting HREC, have previously been identified as a weakness of large scale centralized contact tracing. A UK study reported that two-thirds of index cases did not fully participate [26]. A US study, with coverage and average HREC numbers comparable to Belgium's contact tracing [24], identified unwillingness to participate in contact tracing and more specifically unwillingness to report contacts as a likely cause for low secondary coverage [27]. Across US jurisdictions, the proportion of cases not naming contacts has been estimated ranging from 35 to 83%. Community engagement has been identified as critical for the successful implementation of, and public adherence to, contact tracing [28]. Contact tracing's voluntary steps (seeking testing and interacting with tracers) have been reported to determine the efficiency [29].

Since the actual number of HREC is unknown, we cannot link our observed average number of HREC to any statement on reporting behavior by index cases. The reported number however was, even given the measures in place specifically to avoid HREC, low and relatively stable over time. Other sources on contact patterns during the study period in Belgium such as the CoMix survey and the 'great corona survey' [30, 31] have registered

a higher number of contacts. We observed some indications of selective reporting: e.g. index cases with symptoms were more likely to report HREC. In previous work, we found that if fewer HREC were reported, the reported HREC were associated with higher test positivity, indicating that typically the highest risk HREC were identified by the index case. This observation is linked to the high prevalence of household members among HREC: 63% of HREC were household members, a known particularly susceptible population [32, 33]. Also in our data, observed positivity rates were highest among household contacts [11].

Few other studies have further investigated selective reporting. In Geneva, Switzerland, contact tracing was linked to the overall case database and a residence registry to try and identify the proportion of household clusters that were identified during contact tracing [34]. A lower probability of inclusion in contact tracing was associated with: higher wealth, larger buildings (more residents) and being unvaccinated. Bonacci et al. report that the elicitation of contacts can be improved by reducing the workload and increasing the experience of contact tracers [35].

Household members and the association of several of contact tracing's events with age resulted in the distinctive profile of CIAH. The 6–18y age group was associated with successful contact and reporting of HREC. They and the other age groups present in their household were more prevalent among CIAH. This is illustrated in the bimodal age density curve for CIAH with peaks at 12 and 40 years of age (Supplementary Fig. 1). When interpreting overall case statistics, one should consider the influence of contact tracing which was influenced by design elements, social contact patterns, household structure and the willingness to participate and report.

Finally, with respect to reporting HREC, index cases might inform their close-contacts themselves. This is typically referred to as informal contact tracing and can complement and speed up contact tracing. We could only observe this indirectly: 9% of asymptomatic HREC having gone for a test one day prior to being contacted. Another study reported that the number of persons that requested a PCR-test after a self-reported high-risk contact was much higher than the number included in the centralized contact tracing [15]. A US survey reported that a high proportion of cases will self-notify their HREC, but that quarantining was lower for self-notified HREC [27].

Interrupting transmission requires contact tracing to catch-up to ongoing transmission. Speed should be prioritized over proportions contacted [36]. Infectors who were confirmed ≥ 4 days after symptom onset were associated with higher infectiousness [13] and self-isolation after symptom onset was only half as efficient as isolation

before [37]. We observed that for 30% of primary cases and 29% of CIAH the interval from symptom-onset to test result was 4 days or longer. The long interval for CIAH was remarkable since CIAH were encouraged to get tested once symptomatic. This might be because contact tracing failed to contact 43% of CIAH before symptom-onset. In Finland, HREC were contacted, on average 0.8 days before symptom-onset. This faster contact was a consequence of the overall shorter delays associated with the primary index (faster testing after symptom-onset, faster contact, ...) [37].

Given that the symptom-onset to testing delay was only slightly shorter for CIAH compared to primary cases, despite recommendations and increased awareness, system delays might be an appropriate target for faster contact tracing. To this end, Rapid Antigen-tests (RAT) have been proposed as an alternative to PCR-tests [36], but PCR-tests were preferred because of their high sensitivity, specificity and the central collection of results. Given the high risk of false negative tests, RAT were discouraged for symptomatic persons and HREC [38]. Evaluations in the UK and Netherlands also found longer time intervals to be causal to the limited impact of manual contact tracing [39, 40]. Part of the appeal of digital contact tracing is the reduction of delays. Both Belgium and the Netherlands have however reported longer delays associated with digital contact tracing [40, 41]. Alternatively, improving the speed by design could be done by immediately testing contacts of contacts. Over the study period however, the average HREC tested negative, therefore the majority of contacts of HREC would have been tested without identified exposure.

Our research is subject to limitations, some of which stem from the available data on contact tracing. Due to the scope of the programmatic data collected, we were unable to gauge willingness to participate and report. This limitation is particularly evident in our inability to differentiate between system failures and population adherence. The failure to successfully contact an index case could result from either the call center's inability to reach out or the index case's unwillingness to participate. The association between being vaccinated as index case and reporting HREC could be due to having more contacts or increased compliance to contact tracing requests by vaccinated persons.

This limitation is even greater for HREC, as most remained unidentified until successful contact. Although some could be identified by linking contact tracing codes to laboratory test results, laboratories did not systematically report these codes. Consequently, a portion of the HREC remained unidentified, leading to an underestimation of secondary coverage. Some HREC were contacted via text messages, a method increasingly used towards the end of the study period and during periods of high

incidence. As no response was required to these text messages, no reactions were recorded. The use of text messages and the general issue of identifiability of HREC hindered our ability to ascertain successful contact with HREC. Therefore we did not include successful contact with HREC as a separate event in this paper, but opted for the better-defined 'identifiability'. Unfortunately, this also restricted our ability to compare text messages with telephone interviews as means of contact.

Our inability to include contact tracing details from the separate systems available in collectivities, schools and firms also artificially reduced secondary coverage. Index cases reported collectivities to centralized contact tracing. These collectivities were contacted by the call centers, but any further steps were not documented in the centralized contact tracing system (e.g. how many HREC were identified). Once positive, HREC from collectivities were included as index cases in centralized contact tracing again. The rationale behind separate contact tracing for collectivities was that medical professionals associated with the collectivity were best placed to identify HREC. Japan's contact tracing reported that transmission to hospitals, care facilities, schools and family members were generally located at the edge of transmission chains and spillover transmission from those institutions were mostly household transmission [42]. Unfortunately we were not able to investigate such hypotheses. In addition to the absence of information on collectivities, we also lacked the data to examine contact tracing effectiveness across population subgroups such as urban versus rural, ethnic communities, varying socioeconomic groups, Such granular analysis could reveal disparities in program reach and effectiveness that may inform more targeted and equitable contact tracing strategies in future pandemic responses.

In 2020, contact tracing was less performant (50% of CIAH were contacted while already symptomatic, 1 day delay from test to test result, lower inclusion of cases, ...). We did not further explore these differences in relation to other epidemiological metrics (e.g. hospitalizations). The 20-day time period that was considered for linking cases has no formal justification and could be considered long. This interval has been used in our previous work.

A complete evaluation of contact tracing's performance would also require an investigation into isolation and quarantine compliance of index cases and HREC. Isolation of index cases is considered most important [37], but low adherence has been reported: e.g. 43% in the UK [43]. Given the importance, studies have been designed specifically to investigate alternative approaches. Regular testing of contacts instead of quarantine would be non-inferior [18, 44]. As these and several other aspects remained unobserved (e.g. exact dates of isolation and quarantine, informal contact tracing), contact tracing's

effectiveness can only be quantified under untestable assumptions. Additional research into how putative transmission chains established through contact tracing translate to actual transmission, would further help evaluate contact tracing's effectiveness. In this paper we included the data as reported from which we established putative transmission chains, acknowledging that transmission source and directionality cannot be definitively assessed from our contact tracing data.

Conclusion

In addition to indicators that illustrate performant contact tracing, i.e. lower test positivity for contacts of CIAH, fewer contacts and contracted serial intervals, this work also identified weaknesses in the design and application of centralized contact tracing, i.e. artificial end points (failure to contact index cases and identify contacts) and considerable delays. Besides a scalable and fast contact tracing system, population participation and adherence are crucial towards more performant contact tracing.

Abbreviations

AG	Antigen
AIC	Akaike's Information Criterion
AVIQ	Agence pour une Vie de Qualité
CI	Confidence Interval
CIAH	Cases Identified As HREC
COVID-19	Coronavirus disease 2019
GDPR	General Data Protection Regulation
HREC	High-Risk Exposure Contact
INSZ	Identificatienummer van de Sociale Zekerheid (Belgian national registry number)
NPI	Non-pharmaceutical interventions
OR	Odds Ratio
PCR	Polymerase chain reaction
RAT	Rapid Antigen Tests
RNA	Ribonucleic acid
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2

Supplementary Information

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Supplementary Material 1.

Authors' contributions

TB data management, analysis, writing the article, LC design and follow-up of contact tracing, writing the article, DvC design and follow-up of contact tracing, writing the article, JvL writing the article, CK writing the article, revision, NH writing the article, revision, AR supervision, writing the article, revision, AT supervision, writing the article, revision, KB supervision, writing the article, revision, NH supervision, writing the article, revision.

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Data availability

Data on contact tracing and laboratory testing as used in this paper are not publicly available. Anonymized and aggregated datasets from the laboratory result datasets on COVID-19 are available on the Health Data Agency website

(<https://catalog.hda.belgium.be/>). Some datasets are instantly available, for others, access has to be requested. The procedure is explained on the website.

Declarations

Ethics approval and consent to participate

Data linkage and collection within the data-warehouse has been approved by the Information Security Committee, Social Security and Health (reference number IVC/KSZG/21/034) and the Medical Ethics Committee UZ Brussels–VUB on 3 February 2021 (reference number 2020/523). The study was conducted in accordance with the Declaration of Helsinki.

Informed consent was waived based on art. 6 and 9 of the GDPR. The collection is allowed based on general interest (art. 6 GDPR) and regarding article 9 § 2 of the GDPR: processing is necessary for reasons of public interest in the area of public health, such as protecting against serious cross-border threats to health, or ensuring high standards of quality and safety of health care and of medicinal products or medical devices, on the basis of Union or Member State law which provides for suitable and specific measures to safeguard the rights and freedoms of the data subject, in particular, professional secrecy.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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